Who Gets to Live in Productive Cities? Family Transfers, Housing Costs, and Spatial Sorting

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Motivation

High housing costs can restrict access to productive cities.

- Productive cities offer higher wages and better labor-market opportunities, but also require higher housing expenditure.
- Young households have little accumulated wealth. Large down payments can make home purchase difficult even when current earnings are relatively high.
- Parental support can bridge this financing gap.

Research question

How does intergenerational transfer shape location and tenure choice of the young generation and by how much?

- **Location or tenure?** Do intergenerational transfers primarily expand access to productive cities or promote homeownership within them?
- **Determinants of parental transfers.** Which young households receive more parental support, and which parents provide more?
- **Macro consequences.** How do transfers affect equilibrium housing prices and the spatial sorting of workers with different talent levels?

Framework

- **Evidence from urban China.** Household survey data document the relationship between parental housing support and young households' homeownership and location choices.
- **A spatial model with family transfers.** In a calibrated spatial equilibrium model with heterogeneous agents, parents choose how much to give to their adult children, and young households choose where to live and whether to rent or own.
- **Quantify the role of family finance.** Comparisons with and without parental transfers quantify their effects. Policy experiments assess how measures to make housing more affordable interact with family support.

Main findings

Compared with the equilibrium without inter vivos transfers:

- **Homeownership responds much more than residence.** Among ages 20–29, A residence rises by 3.5%, while owning in A rises by 46.0%. About 91% of the ownership response is the conditional-tenure margin.
- **The output gain is modest: 0.23%.** Rental access limits relocation. An inflow of lower-productivity workers reduces average productivity in A, partly offsetting the gain from a larger workforce.
- **Transfers have contrasting effects on housing inequality.** Ownership gaps narrow by children's talent, but widen by parental wealth. Public credit and rental support substitute for family finance at different margins.

Related literature

1. **Housing and spatial allocation**
   Hsieh and Moretti (2019); Favilukis, Mabille and Van Nieuwerburgh (2023); Garriga et al. (2023); Mabille (2023).

2. **Family transfers and homeownership**
   Engelhardt and Mayer (1998); Guiso and Jappelli (2002); Lee et al. (2020); Brandsaas (2025).

3. **Intergenerational resources and inequality**
   Nishiyama (2002); De Nardi (2004); Adermon, Lindahl and Waldenström (2018); Daysal, Lovenheim and Wasser (2023).

Outline

1. Empirical facts
2. Model
3. Baseline
4. Counterfactual
5. Policies

Data and regional groups

- **CHIP 2018: parental housing assistance.**
  2,954 urban households with heads aged 20–39. The housing module asks about parental help when buying or building a dwelling. Current tenure and residence are also recorded.

- **Two regional groups.**
  Sampled cities are divided into two groups.
  - **Region A:** The 13 Tier 1 and Tier 2 cities covered by CHIP, characterized by higher productivity and higher housing costs.
  - **Region B:** The remaining sampled urban cities, which generally have lower productivity and more affordable housing.

Parental housing assistance is common

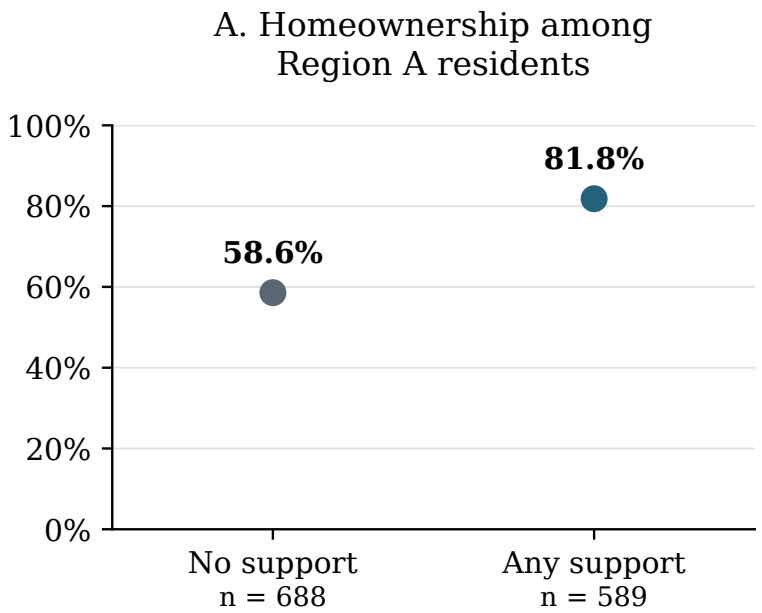
Family support is a substantial source of housing finance.

Sample	Households	Recipients	Share
All young households	2,954	1,458	49.4%
Current homeowners	2,351	1,285	54.7%

- Roughly half of young urban households report parental help with a housing purchase or construction.

Fact 1: Homeownership within productive cities

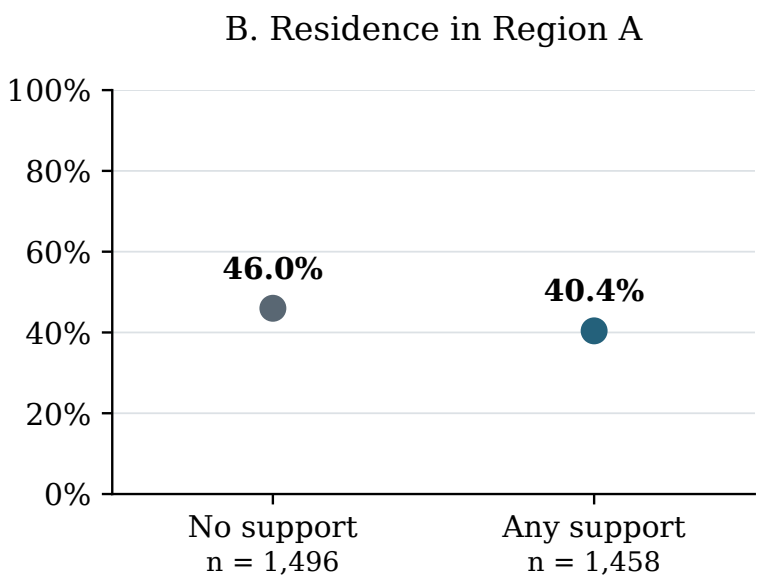
Among households already residing in Region A:



- Ownership is 81.8% among recipients, versus 58.6% among nonrecipients, a gap of 23.3 percentage points.
- Nonrecipients often access A through **renting**. Family finance is strongly associated with tenure within A.

Fact 2: Residence in productive cities

Now use all households in each assistance group as the denominator.



- A residence is 40.4% among recipients, versus 46.0% among nonrecipients, a gap of −5.6 percentage points.
- Higher ownership within A does not imply that recipients are more concentrated in A.

Why a model is needed

Recipient differences do not identify the effect of family transfers.

- **Selection into support.** Parents may give more to children with weaker earning capacity, who also gain less from A's wage premium.
- **Competing household responses.** Support can finance ownership in A without a move, or make ownership in cheaper B attractive relative to renting in A.
- **General-equilibrium adjustment.** Transfer-financed demand changes purchase prices and rents, affecting households with little parental support as well.

The model evaluates these mechanisms jointly and compares **all households across economies** to give a comprehensive answer.

Model environment

- **Two representative regions with different earning opportunities and housing costs.**
  - Region A: higher productivity and expensive housing.
  - Region B: lower earnings and more affordable homeownership.
- **Households differ in talent and family resources.** Permanent talent, parental origin, liquid assets, housing and mortgage debt determine their opportunities.
- **Location and tenure are joint choices.** Rent or own in either region; incumbent owners may keep their house.

Life cycle and intergenerational timing

Six ten-year stages, ages 20–79. Households work through ages 60–69 and retire at 70–79.

Age	Main family event
20–29	Receive parental transfer f; enter labor and housing markets.
50–59	Receive inheritance; observe the child's talent and choose a transfer to the age-20 child.
70–79	Receive a pension; leave a terminal net estate at the age-80 boundary.

Preferences and lifetime earnings

Households derive utility from consumption and housing services.

$$\mathcal{C}(c,h) = \left[(1-\alpha_h)c^{\epsilon} + \alpha_h h^{\epsilon}\right]^{1/\epsilon},$$
$$u^g(c,h,j) = \frac{\mathcal{C}(c,h)^{1-\gamma}}{1-\gamma} + \xi_j^g.$$

- c: consumption; h: housing services; j: location–tenure alternative. The age-specific ξ_j^g capture residual nonpecuniary attractiveness, with $\xi_{rB}^g = 0$.
- **Talent is fixed over the life cycle.** Gross earnings are

$$y_\ell^g(z) = w\,\mathrm{TFP}_\ell G_g z, \qquad \mathrm{TFP}_A > \mathrm{TFP}_B = 1.$$

G_g is age efficiency. Higher talent raises the absolute earnings gain from working in A.

Location and tenure choices

At entry, households choose the alternative with the highest lifetime value.

$$V^1 = \max\Big\{\underbrace{V^1_{rA} + \nu_{rA}}_{\text{Rent in A}},\ \underbrace{V^1_{oA} + \nu_{oA}}_{\text{Buy in A}},\ \underbrace{V^1_{rB} + \nu_{rB}}_{\text{Rent in B}},\ \underbrace{V^1_{oB} + \nu_{oB}}_{\text{Buy in B}}\Big\}.$$

Each V^1_j is the lifetime value of an alternative. The taste shocks ν_j are independent Type-I extreme-value draws.

Young household: renting

An entrant begins with talent z, transfer f, and origin ℓ^{org}. It initially has no liquid assets, house or mortgage debt.

$$\underbrace{c + x'}_{\text{consumption and saving}} + \underbrace{R_\ell h}_{\text{rent}} + \underbrace{\kappa^1(r\ell; \ell^{\text{org}})}_{\text{location costs}} = \underbrace{(1-\tau_l) y_\ell^1(z) + T}_{\text{disposable income}} + f.$$

$$x' \geq 0, \qquad h \geq \underline{h}.$$

- R_ℓ is rent per square meter over the model period; T is the universal government transfer.
- Entry from B to A incurs a resource cost. Renting in A also entails a recurring friction associated with local services.
- **Renting permits productive-city residence without a down payment.** Transfers can improve housing and saving without changing location.

Young household: buying and mortgage finance

Purchasing requires equity and current debt service:

$$c + x' + \underbrace{P_\ell h - M}_{\text{down payment}} + \underbrace{\Pi_{K_1}(M)}_{\text{mortgage payment}} + \kappa^1(o\ell; \ell^{\text{org}}) = Y_\ell^{1,\text{disp}}(z) + f,$$

$$0 \le M \le \theta_{\text{LTV}} P_\ell h,$$
$$\Pi_{K_1}(M) \le \theta_{\text{PTI}} y_\ell^1(z).$$

- M is the new mortgage; K_1 its term; $\Pi_{K_1}(M)$ is the current ten-year payment. Saving remains nonnegative.
- **Transfers relax the cash constraint**, but do not enter the gross-income base of the payment-to-income limit.
- A transfer may finance ownership in B while remaining insufficient for the preferred purchase in A.

Parents choose transfers endogenously

At ages 50–59, the parent knows the child's talent z_c. It chooses $f \geq 0$ jointly with its own location, housing and saving.

$$v_j^4(s^4) = \max_{\boldsymbol{a}_j, f \geq 0} \Big\{ u^4(c, h, j) + \beta \mathbb{E}[\overline{V}^5(s^5)] + \alpha_p \underbrace{\overline{V}^1(z_c, f, \ell_-)}_{\text{child's optimized expected lifetime utility}} \Big\}.$$

Example of a parent keeping its current home

$$c + x' + \underbrace{\Pi_q(m)}_{\text{mortgage payment}} + \underbrace{f}_{\text{transfer}} = Y_{\ell_-}^{4,\text{disp}}(z) + R_s x, \qquad x' \geq 0.$$

The parental giving decision

- $\boldsymbol{a}_j$ collects consumption c, saving x', housing size h (renting or buying), and mortgage borrowing M (buying).
- For the parent keeping its home, liquid assets x already include inheritance, and R_s is their gross return. The current mortgage payment is predetermined, so giving f competes with consumption and saving.
- α_p measures parental altruism by weighting the child's expected lifetime utility. The parent anticipates the child's choices and averages over its taste shocks.
- **Giving reflects the child's needs and the parent's capacity.** Recipients and nonrecipients are selected groups.

Retirement and terminal bequests

- At ages 70–79, households retire and receive pension income $y^{\text{pen}}(z) = \rho_{\text{pen}} w G_4 z$, where ρ_{pen} is the pension replacement rate.

$$v_j^6(s^6) = \max_{\boldsymbol{a}_j \in \Gamma_j^6(s^6)} \left\{ u^6(c, h, j) + \underbrace{\mathcal{B}(e)}_{\text{bequest utility}} \right\}.$$

- Pensions do not depend on region; new mortgage borrowing is unavailable.
- $\mathcal{B}(e)$ is warm-glow utility from the estate e, comprising liquid saving and net housing wealth. These estates generate inheritances at ages 50–59.

Housing supply and tenure conversion

- **Higher purchase prices encourage new construction**, expanding the region's housing stock.
- **Rental and owner-occupied housing are partially segmented**, with limited conversion between them.
- Higher rents relative to purchase prices shift floor space toward rentals. Higher purchase prices relative to rents shift it toward owner occupation.

Stationary spatial equilibrium

Owner and rental housing markets clear in each region.

$$\sum_{g=1}^{6} \int h \, d\mu_{o\ell}^{g} = (1 - s_{\ell}^{r}) H_{\ell},$$

$$\sum_{g=1}^{6} \int h \, d\mu_{r\ell}^{g} = s_{\ell}^{r} H_{\ell}, \qquad \ell \in \{A, B\}.$$

H_ℓ is total housing floor space in region ℓ; s_ℓ^r is its rental share.

- **Government budget balance.** Labor taxes finance pensions and universal transfers; T adjusts to balance the budget.
- **Stationary distributions.** Household choices reproduce the state distribution; inheritances match the distribution of terminal net estates.

Calibration to urban China

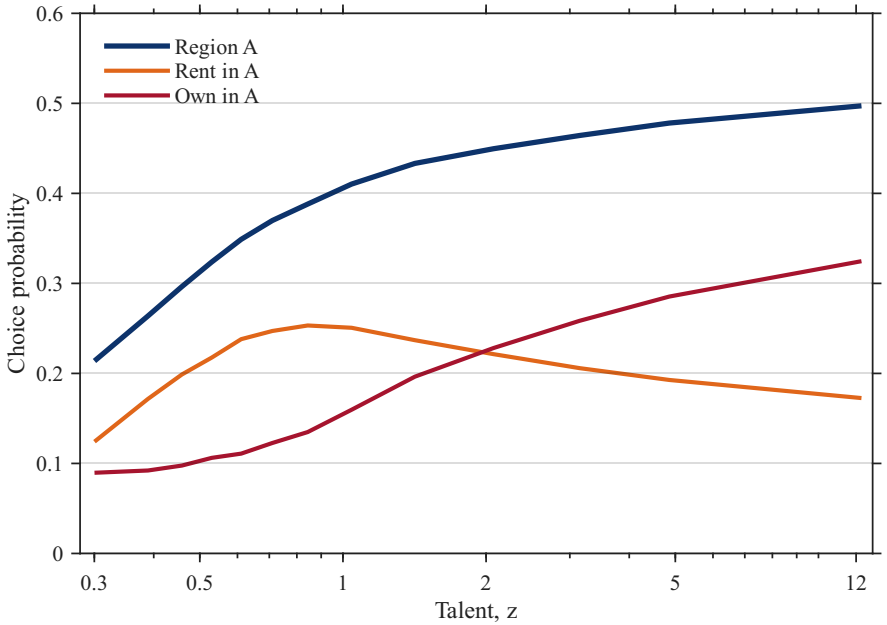
CHFS 2017: earnings, housing and age-specific allocation.
CHIP 2018: parental housing assistance.

Selected targeted moment	Data / target	Model
Receipt among owners aged 20–29 (%)	**62.79**	**62.37**
Working-age A/B income ratio	2.000	1.968
Young *rA–oB* talent gap (%)	9.0	11.4
Purchase price in A (RMB/m^2)	18,370	18,370
Purchase price in B (RMB/m^2)	4,750	4,750
Ten-year rent in A (RMB/m^2)	3,110	3,110
Ten-year rent in B (RMB/m^2)	1,580	1,580

- **Location–tenure shares at six ages** discipline 18 choice constants; homeowner receipt disciplines $\alpha_p = 0.4908$.
- **Purchase prices and rents** discipline construction shifters and the owner–rental allocation of housing supply.

Baseline: talent and location–tenure choices

Choice probabilities among households aged 20–29, conditional on talent.

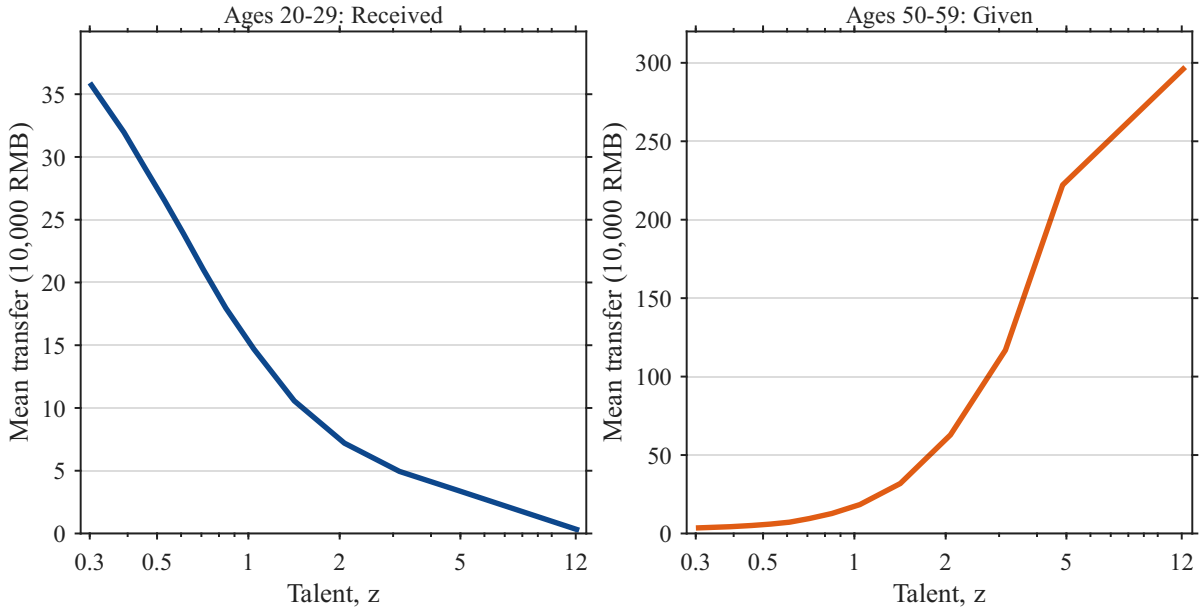


Mean talent	
A owners	1.391
A renters	1.012
B owners	0.908

- Higher talent increases both A residence and A ownership.
- **Renting is most prevalent at intermediate talent.** A renters have higher mean talent than B owners, so access to the productive labor market does not require homeownership.

Endogenous transfers: need and capacity to give

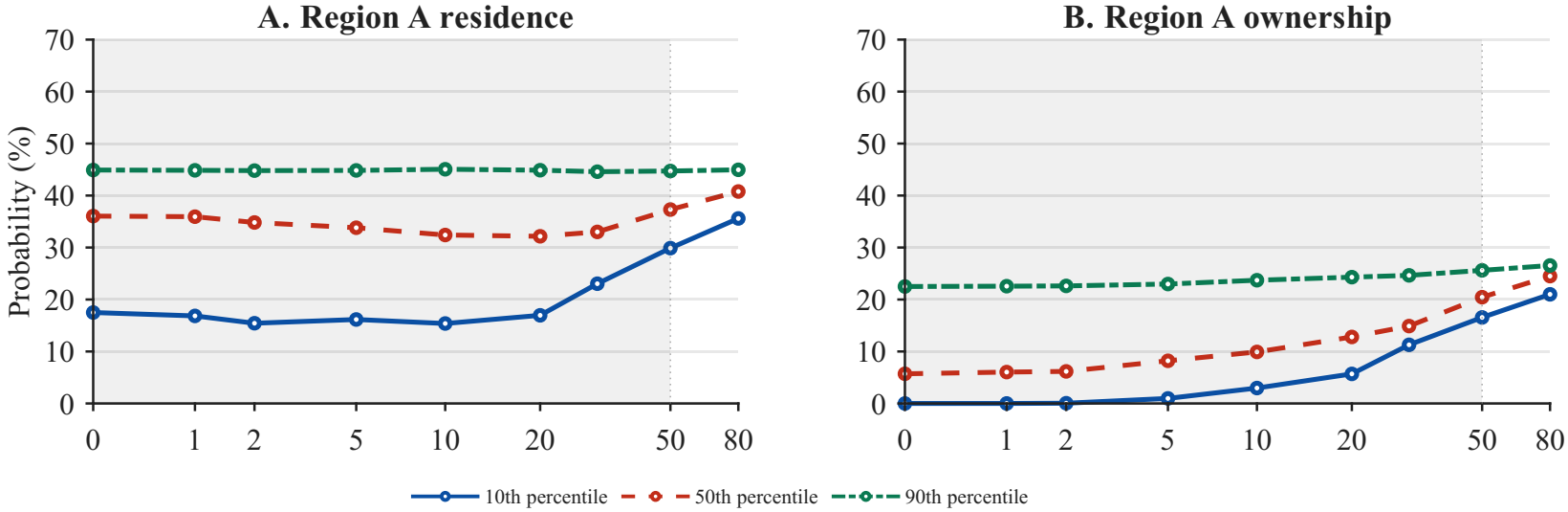
Mean support by child talent (left) and parent talent (right), including zero transfers. Transfers are measured in RMB 10,000.



- Lower-talent children receive more; higher-talent parents give more. The top 1% of parent talent supplies 13.7% of total transfers.
- **58.0% of young households receive support.** The mean across all young households is RMB 217,000; receipt and transfer size both contribute to these profiles.

Transfers finance A ownership more than A residence

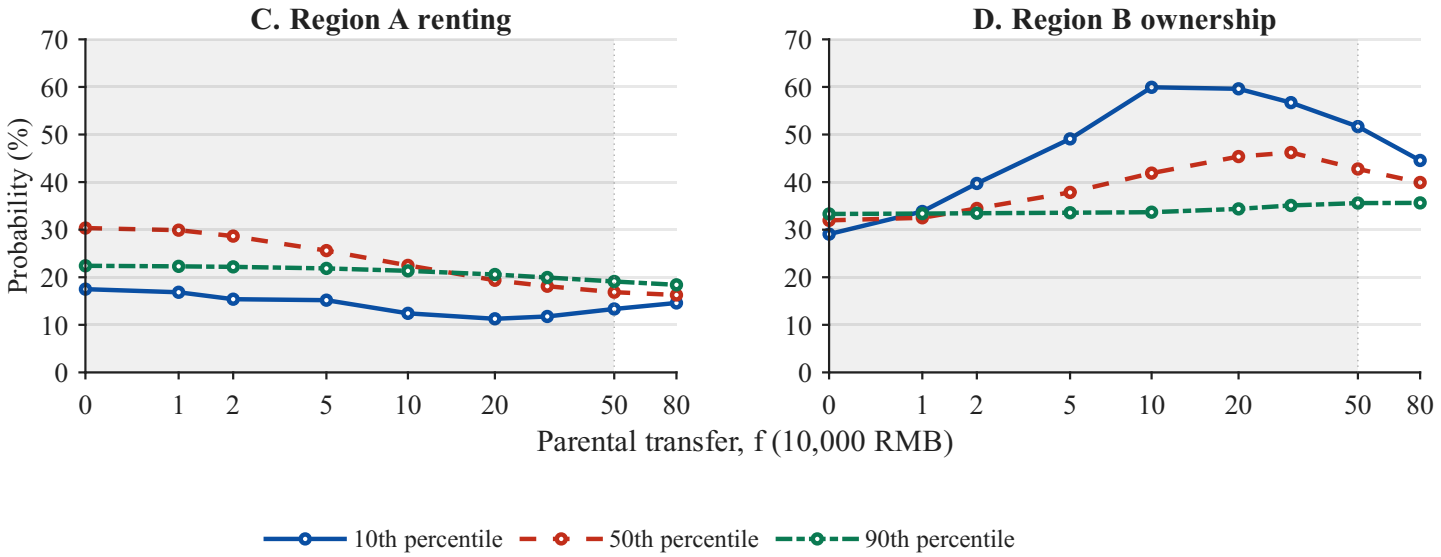
Vary transfers (RMB 10,000) at fixed talent and economic environment.



- **Transfers ease A's upfront purchase burden**, especially for low- and middle-talent households.
- **At high talent, A residence remains nearly unchanged as ownership rises.** The corresponding decline in A renting shows that transfers mainly change tenure within the productive region.
- **Modest transfers can lower A residence at lower talent** by favoring cheaper B ownership. Larger transfers make A affordable.

Modest transfers can encourage ownership in B

The same transfer experiment also changes the attractiveness of renting in A and purchasing in the less expensive region.



- At low talent, raising support from zero to **RMB 20,000** increases B ownership from 29.1% to 39.7%; A ownership remains near zero.
- At intermediate talent, declining A renting initially outweighs rising A ownership. Thus a transfer can improve housing tenure while reducing productive-city residence.

Family finance relative to young buyers' needs

Baseline homeowners aged 20–29. Monetary values below are RMB 10,000.

	Owners in A	Owners in B
Mean purchase value	70.35	23.59
Down payment / ten-year income	50.1%	42.4%
Mean origination LTV	44.1%	32.9%
Transfer / down payment	**85.1%**	**140.0%**

- Buying in A entails a much larger purchase value and a larger down-payment burden relative to the buyer's own earnings.
- Transfers provide substantial purchasing resources. LTV averages individual buyer ratios; the other ratios use group totals.

What changes when the transfer channel is shut down?

Set the age-4-to-age-1 transfer to zero in every state. Parents remain altruistic. Re-solve housing prices, rents, choices, and the inheritance distribution.

Young-household share (%)	B^0: no transfer	B: transfer	$B - B^0$
Reside in A	33.92	35.10	+1.18 pp
Own in A	**9.33**	**13.63**	+4.30 **pp**

- Transfers increase A residence by **3.5%**, but A ownership by **46.0%**, relative to the no-transfer economy.
- Both shares use all households aged 20–29 as the denominator.

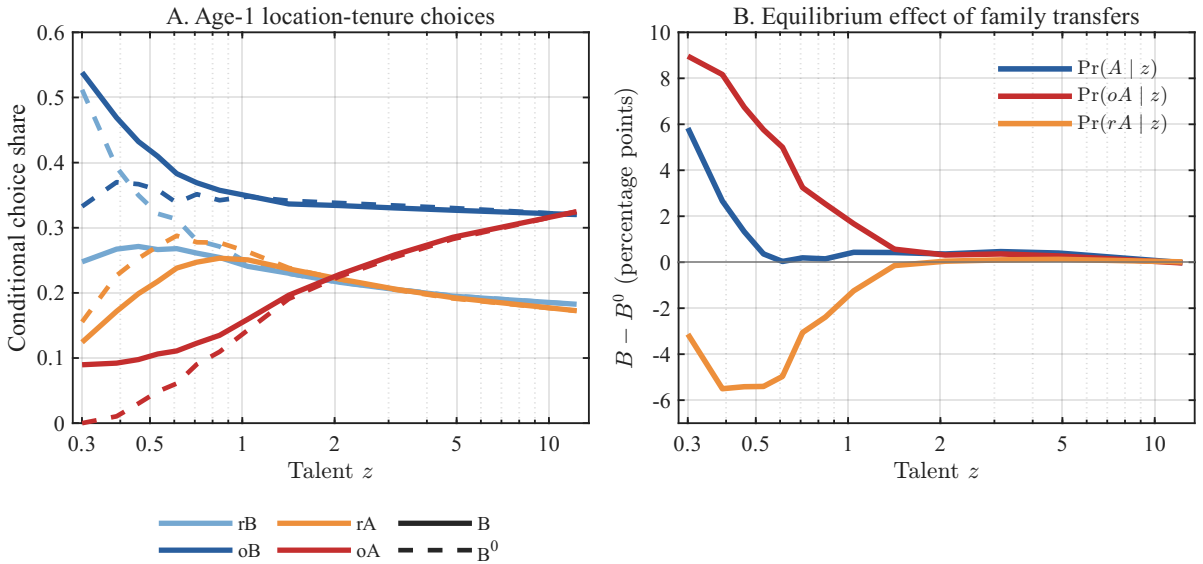
The adjustment is mainly from renting to owning

Location–tenure shares among all households aged 20–29, in percent.

	No transfers	Baseline	Difference (pp)
Rent in B	31.24	25.14	−6.10
Own in B	34.84	39.75	+4.91
Rent in A	24.59	21.47	−3.12
Own in A	**9.33**	**13.63**	+4.30
Own in either region	**44.17**	**53.38**	+9.21

- Family finance expands ownership in **both regions**. The decline in A renting offsets much of the increase in A ownership.
- Cheaper B ownership is an important alternative to A renting.
- **About 91% of the increase in young households' A ownership** comes from higher ownership conditional on A residence.

Who gains ownership in the productive region?



- **Transfers ease purchase financing for lower earners.** The bottom 80% of talent account for 98.0% of the A-ownership expansion.
- **Transfers enable some households who would otherwise rent** to choose ownership in the same region, in both A and B.

Housing prices rise, but buyers rely less on debt

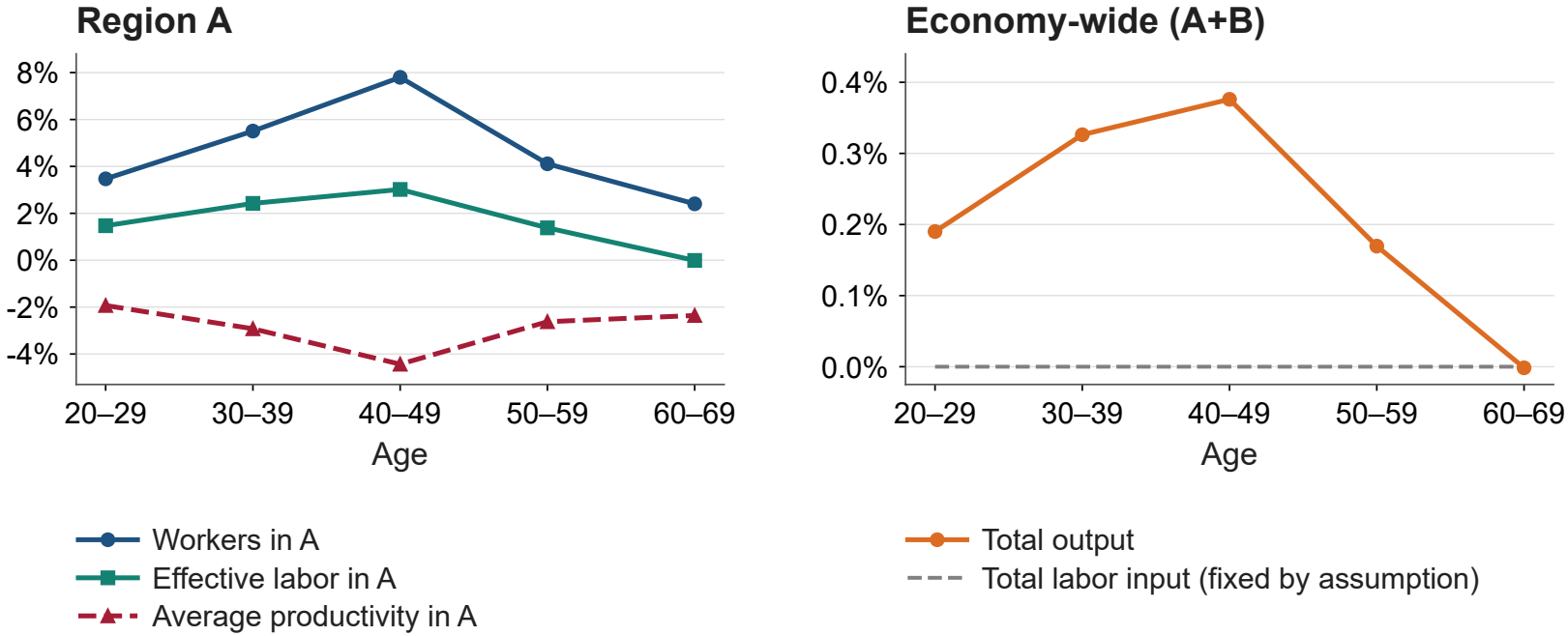
Market levels and buyer leverage, with percentage changes relative to B^0.

	No transfers	Baseline	Effect
Purchase price A	1.757	1.837	+4.55%
Purchase price B	0.436	0.475	+8.98%
Young A-buyer LTV	**68.8%**	**44.1%**	−24.7 **pp**

- Purchase-price increases do not offset the financing advantage of parental support. Housing supply expands and shifts toward owner use.
- **Lower-earning A owners also use less leverage.** Family resources finance more equity; buyer statistics include selection. Purchase prices are in RMB 10,000 per m^2.

Why is the aggregate-output gain small?

Changes with transfers relative to no transfers, covering all workers.



- **More workers in A, but lower average productivity.** Across working ages, composition offsets 61.4% of the population contribution.
- **Aggregate labor input is fixed by assumption.** Reallocation to A, with 37% higher TFP, raises output by **0.23%**.

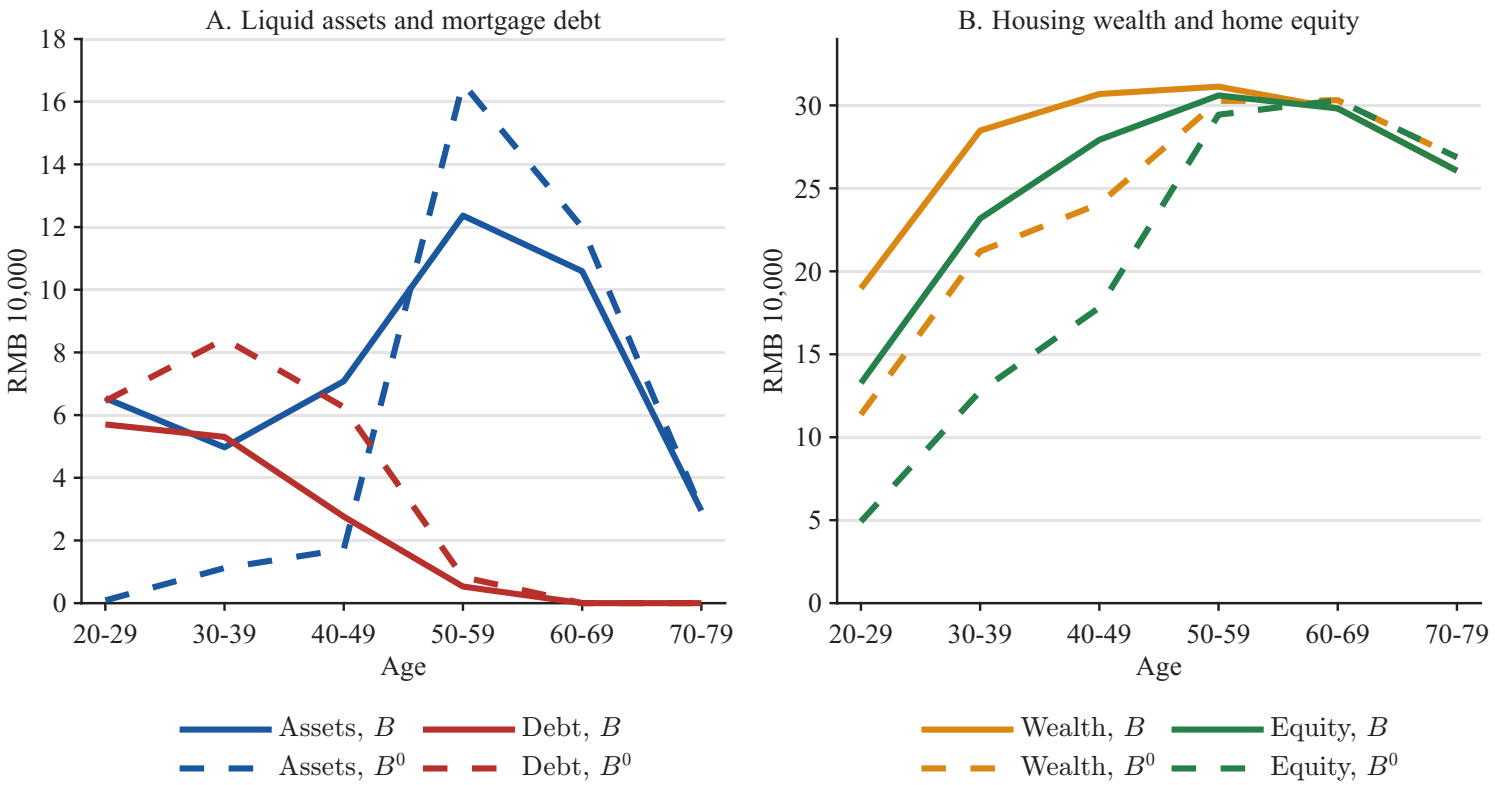
Less dependence on own talent, more on parental wealth

Family transfers change which resources determine young households' ability to own in A.

	No transfers	Baseline	Change
Own-talent ownership gap (pp) Top 20% minus bottom 80%	15.86	11.03	−4.83
Parental-wealth gap (pp) Q4 minus Q1	**0.52**	**10.88**	+10.37
Young home-equity Gini	0.780	0.721	−0.058
Consumption–housing Gini (all ages)	0.386	0.342	−0.044

- Support reduces dependence on own earnings but strengthens the link to family wealth. Equity and consumption–housing inequality decline.

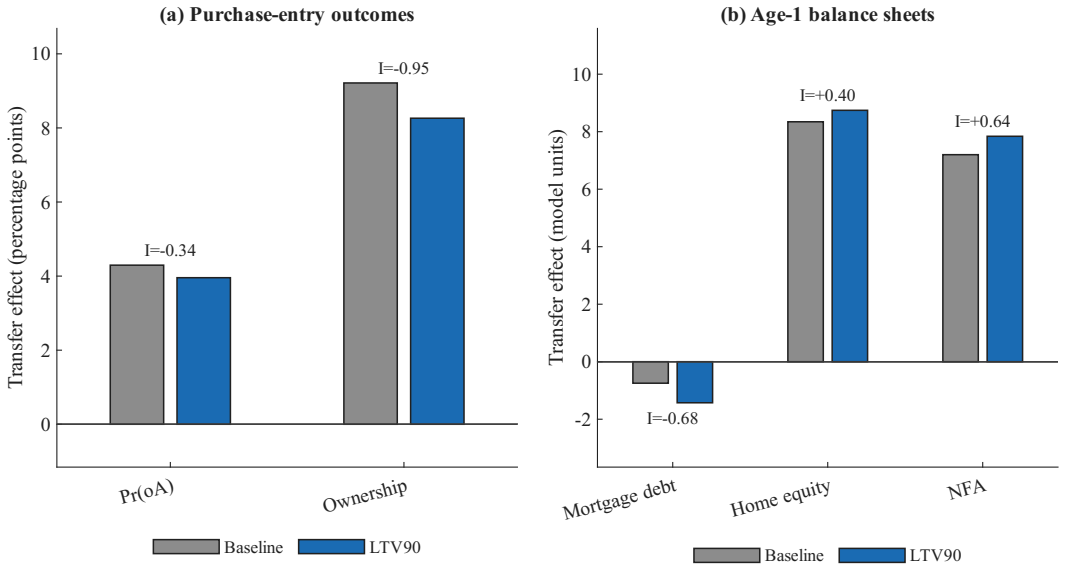
Balance-sheet effects persist; welfare gains differ



- Young equity: RMB 49,200 $\rightarrow$ 132,600. Gains persist at ages 30–49, then narrow when households become donors.
- Consumption–housing CEV: **young 26.33%**; all living 11.60%; young top-5% talent -6.82%. Lower-talent entrants benefit most from family support.

Mortgage credit partly substitutes for family finance

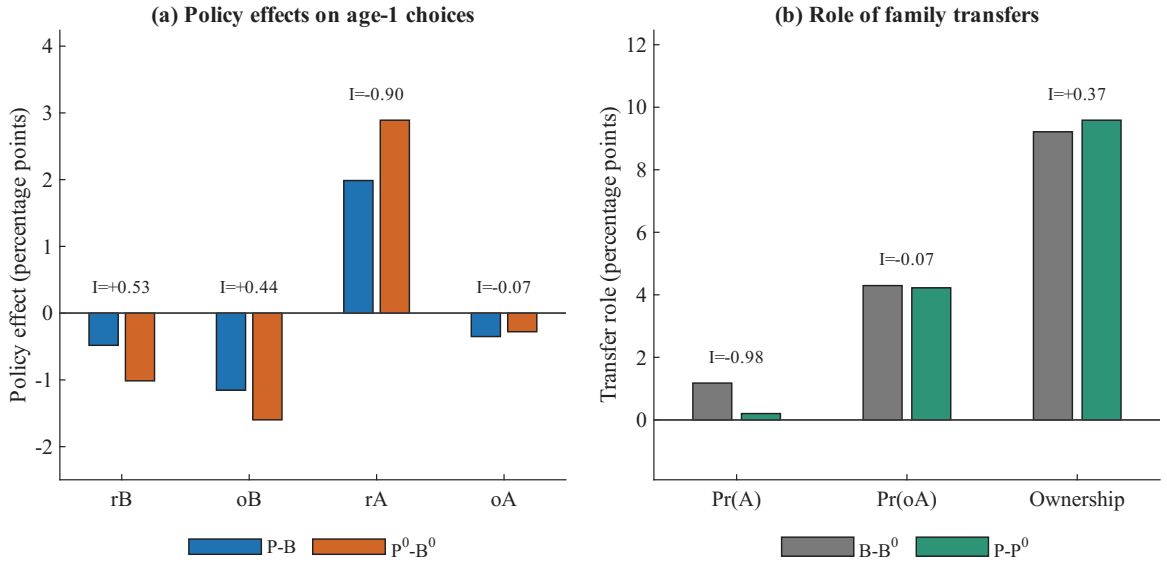
Experiment: LTV ceiling 70% → 90% for new purchases in both regions.



- Young A ownership: +0.13 **pp with transfers**; +0.47 **pp without**. Formal credit partly replaces family finance at purchase entry.
- Bars show transfer effects before and after reform. Transfers still reduce debt and raise equity (all young households; RMB 10,000).

Rental assistance expands access without ownership

Experiment: eligible nonowners aged 20–39 receive a 20% A-rent discount up to 70 m^2.



- Young A residence: +1.64 **pp with transfers**; +2.61 **pp without**. Rental assistance partly replaces family support in productive-city access.
- Ownership falls; transfers retain their A-ownership role. A rent rises 2.99%; spending equals 2.14% of tax revenue.

Conclusion

- **Family transfers expand ownership and reshape sorting.** Young A ownership rises by 4.30 pp, versus 1.18 pp for residence. Rental access preserves productive-city residence; some households buy in lower-cost Region B rather than rent in Region A.
- **The aggregate-output gain is limited.** The residence response is small, and lower average worker productivity partly offsets workforce expansion in A.
- **Family resources reshape housing inequality.** Ownership gaps narrow by children's talent but widen by parental wealth. Credit and rental support act on different affordability constraints.

Housing policy should distinguish access to a productive location from access to homeownership there.

Backup: Spatial and tenure decomposition

Write $A = \Pr(A)$ and $q = \Pr(\text{own} \mid A)$, so $\Pr(oA) = Aq$.

$$\Delta \Pr(oA) = \underbrace{\bar{q}\,\Delta A}_{\text{spatial component}} + \underbrace{\bar{A}\,\Delta q}_{\text{tenure component}} .$$

	Spatial (pp)	Tenure (pp)	Tenure share
Ages 20–29	**0.39**	**3.90**	**90.9%**
Working ages	0.85	1.60	65.2%
All households	0.76	1.15	60.3%

- Among young households, conditional ownership in A rises 11.31 pp; the corresponding residence increase is only 1.18 pp.
- Tenure accounts for most of the response even when later ages are included. Overbars are arithmetic averages across equilibria; the decomposition is exact before rounding.

Backup: Inheritance and family links

- **The explicit family link is an inter vivos transfer.**
  A parent aged 50–59 observes its child's talent and chooses a transfer to the child aged 20–29. The child inherits the parent's initial region.
- **Terminal estates remain in the no-transfer economy.**
  Estates left after ages 70–79 generate the common distribution of inheritances received at age 50. This distribution adjusts in equilibrium.
- **Benchmark independence assumptions.**
  Inheritance is independent of talent and earlier transfer receipt. Parent and child talent are independent in the benchmark; persistence is studied in sensitivity analysis.

Backup: Sensitivity to rental and entry costs

Ownership responds robustly; the spatial response is more sensitive.

Transfer effect, ages 20–29 (pp)	A residence	A ownership
Baseline	+1.177	+4.295
No A-rental friction	−1.231	+4.370
Double A-rental friction	+3.189	+4.048
Half A-entry cost	−0.412	+4.079

- Easier rental access can reverse the location response while preserving the positive ownership response.
- The robust message is the distinction between spatial and ownership access; the sign of the spatial effect is not universal.

Backup: Welfare criterion and heterogeneity

- **Remaining-lifetime self welfare.** Scale consumption and housing services along the fixed no-transfer allocation until it attains baseline welfare. Retain deterministic location–tenure utility and terminal warm glow; exclude idiosyncratic taste utility and the altruistic child value.

- **Transfer-economy gains are heterogeneous.**
  Among ages 20–29: all 26.33%; bottom talent quintile 53.73%; top 5% −6.82%. All living: 11.60%.

- **Policy CEVs use a cost convention.** All-living welfare costs are 0.74% for LTV relaxation and 0.43% for rental assistance: compensation needed in the policy economy to attain its baseline.

Backup: Selected quantitative comparisons

Ages 20–29	No transfers	Baseline
Own in A, bottom 80% talent (%)	6.16	11.42
Own in A, top 20% talent (%)	22.02	22.45
A-buyer origination LTV (%)	68.8	44.1
Liquid assets (RMB 10,000)	0.09	6.54
Mortgage debt (RMB 10,000)	6.45	5.70
Home-equity Gini	0.780	0.721

- More lower-talent households become A owners, while average mortgage leverage falls.
- LTV conditions on A buyers; assets and debt average over all young households, including renters.

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